

Anurag Aggarwal

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EDUCATION

University of Western Ontario

BESc, Mechatronic Systems Engineering & Artificial Intelligence Systems Engineering – Dual Degree.
Cum. GPA: 3.9/4.0

London, Ontario

Sept. 2023 – Present

WORK EXPERIENCE

Ontario Ministry of Health, MPBSDP Branch

Software Quality Assurance Analyst – Co-op

North York, Ontario

May 2026 – Present

- Performed functional and regression testing of web and mobile applications, documenting results and defects to support release readiness.
- Developed and executed **Salesforce** test cases against user stories and acceptance criteria within **Agile/Scrum** sprints.
- Worked with developers and business analysts to reproduce defects, verify fixes, and communicate unresolved issues before production release.

Tethos Association

Embedded Systems and Mechatronics Developer

London, Ontario

Sept. 2025 – Sept. 2026

- Collaborated with mechanical and electrical teammates on sensor-glove casing design, material selection, and circuit prototyping.
- Programmed an **Arduino Uno** to read **gyroscope** and **flex-sensor** inputs and map sensor data to interactive game controls.
- Tested sensor response and hardware–software integration, identifying calibration and prototype stability issues.
- Performed Ergonomical Redesigns on glove casing to support all the electronics in a safe and confined form factor.

Kissan Engineers (Casting)

Industrial Automation & Computer Vision Intern

Batala, Punjab, India

May – Sept. 2025

- Developed an **OpenCV inspection** pipeline using **part segmentation** and **defect classification** to identify **defective** cast components reducing false checks by 62% with supporting operator review.
- Integrated inspection outputs with modified **PLC control** logic for part flagging and coordination with conveyor sensors and actuators.
- Built **Python** and **SQL data workflows** and **Power BI** dashboards to monitor inspection results, defect trends, and production quality metrics.

PROJECTS

SpatialMind | Object-Centric Spatiotemporal Visual Memory System

Python, PyTorch, OpenCV, SAM, DINOv3, FastAPI, PostgreSQL/pgvector

Sept. 2026 – Present

- Developing an **object-centric CV system** combining **open-vocabulary segmentation**, multi-object tracking, and **foundation-model embeddings** to maintain persistent object identities across continuous video.
- Designing a dynamic **spatiotemporal scene graph** that records object locations, spatial relationships, movement, appearances, and disappearances to model how real-world environments change over time.
- Building a **multimodal retrieval and reasoning pipeline** using **vector search** and vision-language models to answer **natural-language queries** about object histories, scene changes, and last-seen events.

Robotic Eraser Collection & Sorting System | ESP32 + Mechatronic System Design

ESP32, Embedded Systems, SolidWorks, Sensors

Jan. – Apr. 2026

- Designed and integrated an **ESP32**-based robot for eraser collection and colour sorting, combining a motorized drive, pickup mechanism, sensors, and actuators.
- Refined the drive and pickup mechanisms through prototype testing, adjusting gear reduction of 4:1 and mechanical clearances to address collection failures.
- Programmed **PWM** motor control and **timed state transitions** to coordinate movement, pickup, sorting, and return sequences.
- Implemented normalized RGB colour classification and actuator control for automated sorting, with **96% colour-sorting accuracy** in controlled prototype trials

Plant Disease Classifier | CNN-Based Computer Vision Classification System

Python, TensorFlow, OpenCV, Raspberry Pi, Embedded C

Sept. – Dec. 2025

- Developed a custom convolutional neural network to classify **38 plant health classes**, achieving **96% accuracy** on a held-out test set from real world examples.
- Integrated **leaf segmentation and frame-level video processing** to isolate individual plant leaves enabling disease identification from continuous video
- Deployed the trained vision pipeline on a **Raspberry Pi**, tested camera-based inference under varied lighting and backgrounds achieving **87% inference accuracy** over prototyping trials.

TECHNICAL SKILLS

Languages: Python, SQL, Embedded C, Java, JavaScript/TypeScript

ML/CV: PyTorch, TensorFlow, OpenCV, CNNs, Object Segmentation, VLMs, LangGraph

Embedded & Automation: ESP32, Arduino, PLC programming, Sensors, I2C, Schematic Interpretation

Backend & Data: FastAPI, PostgreSQL, pgvector, MySQL, PowerBI

Tools & Engineering: Git, Docker, Jira, SolidWorks, Salesforce, testing, Vercel